1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC) tumors that are MSI/dMMR and contain TLS-like lymphoid aggregates, antitumor immunity and response to immune checkpoint blockade (ICI) are governed by tumor-intrinsic pyrimidine metabolism via DPYD. Specifically, high DPYD activity creates a pyrimidine metabolite milieu that supports TLS maturation, dendritic cell antigen presentation, and effector T-cell function, leading to robust ICI responses. In contrast, DPYD downregulation or low activity leads to accumulation of pyrimidine metabolites that suppress TLS maturation and antigen presentation, resulting in poor ICI responses despite high neoantigen burden. Thus, DPYD expression/activity functions as a functional modifier of the TLS–neoantigen axis and predicts ICI outcomes beyond TLS presence or MSI status alone. Testable predictions include: (1) DPYD expression positively correlates with TLS maturity markers and ICI responsiveness in TLS-rich MSI/dMMR MIBC; (2) driving DPYD expression or activity in MSI/dMMR bladder cancer models enhances TLS formation/function and improves ICI efficacy, whereas DPYD inhibition diminishes these features; (3) metabolomic profiling will reveal distinct pyrimidine metabolite signatures that distinguish DPYD-high vs DPYD-low TLS+ tumors in terms of immunologic competence.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSI/dMMR, DPYD
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSI/dMMR: MSI/dMMR and high neoantigen/TMB should sensitize MIBC to immune checkpoint blockade, yet a subset of MSI-high/TMB-high tumors with apparent local immune structures (TLS/TIL) do not respond.

DPYD: DPYD/TP53 metabolic/prognostic axis (offers a tumor-intrinsic metabolic modifier that could blunt immune efficacy).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TP53	Supporting modifier from knowledge graph

Secondary	TMB	Supporting modifier from knowledge graph
Secondary	MSH6	Supporting modifier from knowledge graph
Secondary	SARIFA	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	BCG_immunotherapy_context	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: and T-cell activation; (c) ex vivo metabolomics of pyrimidine metabolites in TLS+ tumors stratified by DPYD; (d) in vivo mouse models (humanized or syngeneic) of MSI/dMMR bladder cancer with DPYD manipulation to evaluate TLS formation, antigen presentation, T-cell infiltration, and response to anti-PD-1/PD-L1; (e) integration with additional modifiers (TP53 status, TMB, MSH6, HDL-C, SARIFA) to assess interaction effects.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.